教材190页8; 213页1,11; 218页2,3.
教材189-190页2,3,4,9,14,15; 219页4(a,b).

190-8

Let $A, B \in F^{n\times n}$. Prove that if $I - AB$ is invertible, then $I - BA$ is invertible and

$$(I - BA)^{-1} = I + B(I - AB)^{-1}A. \tag{1}$$

Proof: Note that

$$(I - BA)(I + B(I - AB)^{-1}A) = I - BA + (I - BA)B(I - AB)^{-1}A = I - BA + BA = I. \tag{2}$$

Hence $I - BA$ is invertible and its inverse is $I + B(I - AB)^{-1}A$.

213-1

Let V be a finite-dimensional vector space and W_1 be any subspace of V. Prove that there is a subspace W_2 of W such that $V = W_1 \oplus W_2$.
Proof: Take a basis $\{\alpha_1, \cdots, \alpha_k\}$ of W_1 and extend it to a basis $\{\alpha_1, \cdots, \alpha_n\}$ of V. Let $W_2 = \mathrm{Span}(\alpha_{k+1}, \cdots, \alpha_n)$ then $V = W_1 + W_2$ and $W_1 \cap W_2 = 0$, so $V = W_1 \oplus W_2$.

213-11

Let V be a vector space, and $W_1, \cdots, W_k$ be subspaces of V. Let $V_j = W_1 + \cdots + W_{j-1} + W_{j+1} + \cdots + W_k$. Show that $V = W_1 \oplus \cdots \oplus W_k$. Suppose that $V = W_1 \oplus \cdots \oplus W_k$. Prove that the dual space V^* has the direct-sum decomposition $V^* = V_1^0 \oplus \cdots \oplus V_k^0$.
Proof: If $f_1 + \cdots + f_k = 0$ for some $f_i \in V_i^0$, then for any $1 \leqslant i \leqslant k$ and $w \in W_i$, $f_j(w) = 0 \forall j \neq i$ so $f_i(w) = 0$. Hence $W_i, V_i \subset \mathrm{Ker} f_i$ so $V = W_i + V_i \subset \mathrm{Ker} f_i$ and $f_i = 0$.
For any $f \in V^*$, let $f_i = f\iota_i\pi_i$ where $\pi_i : V \to W_i$ and $\iota_i : W_i \to V$ are the projection and embeddings. Then $f_i \in V_i^0$ and $f = f_1 + \cdots + f_k$. Hence $V^* = V_1^0 \oplus \cdots \oplus V_k^0$.

218-2

Let T be the linear operator on $\mathbb{R}^2$, the matrix of which in standard ordered basis is $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.
Let W_1 be a subspace of $\mathbb{R}^2$ spanned by $e_1 = (1, 0)$.
(a) Prove that W_1 is invariant under T.
(b) Prove that there is no subspace W_2 which is invariant under T and is complementary to W_1: $\mathbb{R}^2 = W_1 \oplus W_2$.
Proof: (a) $Te_1 = 2e_1$ so W_1 is invariant under T.
(b) If $Tv = \lambda v$ for $v = (a, b)$ then $2a + b = \lambda a, 2b = \lambda b$ so $\lambda = 2$ and $v \in W_1$. Hence if $\mathbb{R}^2 = W_1 \oplus W_2$ then $\dim W_2 = 1$. Suppose $W_2 = \mathrm{Span}(e_2)$ then $Te_2 = \lambda e_2$ implies $e_2 \in W_1$, a contradiction.

218-3

Let T be a linear operator on a finite-dimensional vector space V. Let R be the range of T and N be the null space of T. Prove that R, N are independent iff $V = R \oplus N$.
Proof: Note that $\dim R + \dim N = \dim T$. If R, N are independent, then
$\dim V = \dim R + \dim T = \dim R \oplus N$ so $V = R \oplus N$. The converse is trivial.

189-2

Let V be a $n-$dimensional vector space over F. What is the characteristic polynomials of the identity operator on V? What is the characteristic polynomial for the zero operator?
Solution: $f_{id} = (x - 1_V)^n$, and $f_0 = x^n$.

189-3

Let A be an $n \times n$ triangular matrix over the field F. Prove that the characteristic values of A are the diagonal entries of A, i.e., the scalars $A_{i,i}$.
Proof: The characteristic polynomial $f_A = \prod_{i=1}^n (x - A_{i,i})$. Since every eigenvalue is a root of f_A, it is one of $A_{i,i}$.

189-4

Let T be the linear operator on $\mathbb{R}^3$ which is represented in the standard ordered basis by the matrix

$$\begin{pmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{pmatrix}. \tag{3}$$

Prove that T is diagonalizable by exhibiting a basis for $\mathbb{R}^3$, each vector of which is a characteristic vector of T.
Proof: $(1, 2, 0), (1, 0, 2)$ are eigenvectors of T with eigenvalue -1, and $(1, 1, 2)$ is an eigenvector with eigenvalue 3.

190-9

Use the result of Exercise8 to prove that, if $A, B \in F^{n \times n}$ then AB and BA have precisely the same characteristic values in F.
Proof: If $\lambda \neq 0$, λ is an eigenvalue of AB iff $\lambda I - AB$ is invertible, iff $\lambda I - BA$ is invertible, iff λ is an eigenvalue of BA. If $\lambda = 0$, $-AB$ is invertible iff $-BA$ is invertible, since $\det AB = \det BA$.

190-14

Let A be an $n \times n$ diagonal matrix with characteristic polynomial $(x - c_1)^{d_1} \cdots (x - c_k)^{d_k}$ where $c_1, \cdots, c_k$ are distinct. Let V be the space of $n \times n$ matrices B such that $AB = BA$. Prove that the dimension of V is $d_1^2 + \cdots + d_k^2$.
Proof: Suppose $AB = BA$. For any $1 \leqslant i \leqslant k$, and any $v \in W_i = \mathrm{Ker}(c_i I - A)$,
$ABv = BAv = Bc_i v = c_i Bv$ so $Bv \in W_i$. Hence W_i is also invariant under B, and B is uniquely determined by the operators $B|_{W_i} \in \mathcal{L}(W_i)$.
Conversely, if W_i are invariant under B, then $AB|_{W_i} = BA|_{W_i}$. Since $F^n = W_1 \oplus \cdots \oplus W_n$, $AB = BA$.

Therefore there is a bijection between V and $\mathcal{L}(W_1) \times \cdots \times \mathcal{L}(W_k)$. Hence $\dim V = \sum \dim \mathcal{L}(W_i) = \sum_{i=1}^{k} d_i^2$.

190-15

Let V be the space of $n \times n$ matrices over F. Let A be a fixed $n \times n$ matrix over F. Let T be the operator $L_A \in \mathcal{L}(V)$. Is it true that A, T have the same characteristic values?
Solution: Yes, A is the matrix of T under the standard basis of F.

219-4

Let $T \in \mathcal{L}(V)$ and $V = W_1 \oplus \cdots \oplus W_k$ where each W_i is invariant under T. Let T_i be the induced operator on W_i.
(a) Prove that $\det T = \det T_1 \cdots \det T_k$.
(b) Prove that the characteristic polynomial for f is the product of the characteristic polynomials for $f_1, \cdots, f_k$.
Proof:
(a) Take a basis $\mathcal{B}_i$ for W_i and combine them to obtain a basis $\mathcal{B}$ of V, then $[T]_{\mathcal{B}}$ is block diagonal, so $\det T = \det [T]_{\mathcal{B}} = \det [T_1]_{\mathcal{B}1} \cdots \det [T_k]_{\mathcal{B}_k}$.
(b) Likewise $f_T(c) = \det(cI_n - [T]_{\mathcal{B}}) = \prod_{i=1}^{k} \det(cI_{k_i} - [T_i]_{\mathcal{B}_i}) = f_1(c) \cdots f_k(c)$.